

AI Large Model - Dify

1. Dify Overview

Dify is an open-source LLM application development platform that combines BaaS and LLMOps concepts. It supports low-code/no-code rapid building of production-grade generative AI applications, is compatible with hundreds of models, and has built-in enterprise-level RAG and visual workflows. It supports private deployment and full-cycle LLMOps control.

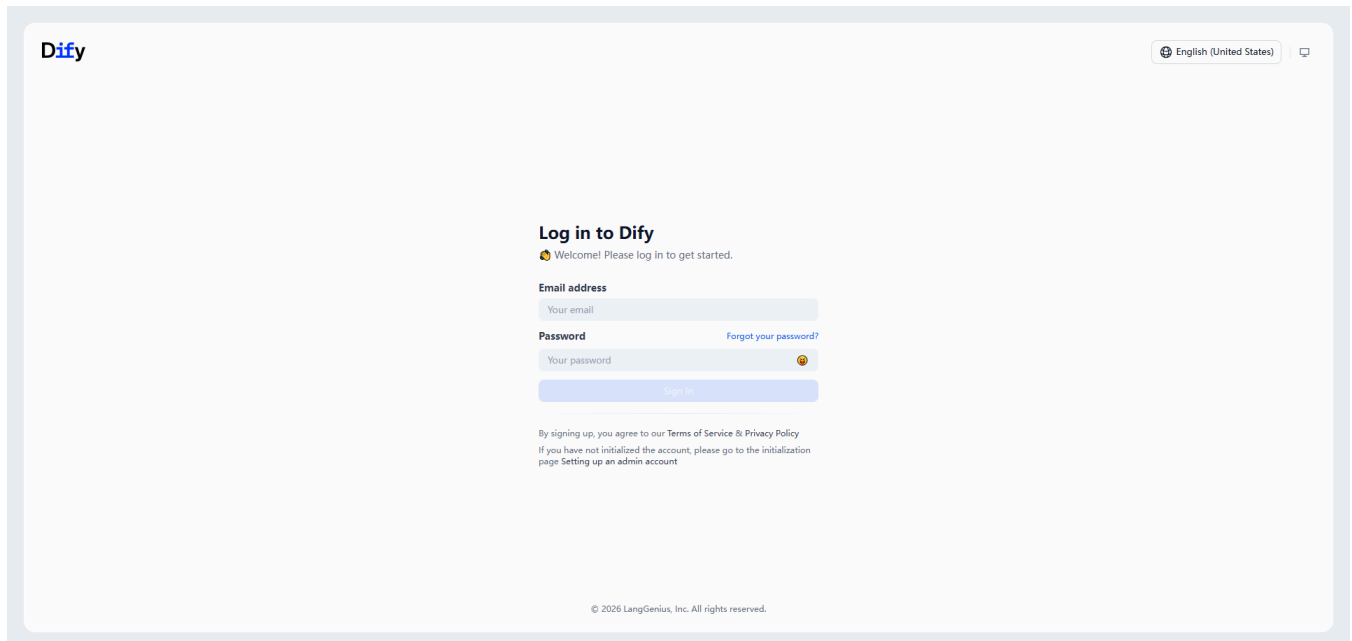
Dify (Define + Modify) is designed for developers and enterprises, providing an integrated solution of "backend as a service + BaaS + LLMOps" to reduce the barrier to transforming generative AI applications from prototype to production while ensuring data security and compliance.

One of Dify's product forms is the "AI Application Factory," which enables "rapid creation of chatbots and content generators through a low-code interface." By configuring the **Decision Layer** and **Execution Layer**, we can plan task steps and select execution function actions to achieve dual-model reasoning. Additionally, for more complex features, by adding a **Knowledge Base**, the **Decision Layer** can reference content from the **Knowledge Base** when planning actions to generate correct steps.

2. Entering the Dofbot-SE Dify Configuration Interface

Dofbot-SE Dify service starts automatically on boot

Enter the virtual machine's IP address in your browser to access the configuration interface. The interface is shown below,



Dify English (United States)

Log in to Dify
👋 Welcome! Please log in to get started.

Email address
Your email

Password [Forgot your password?](#)
Your password

[Log in](#)

By signing up, you agree to our [Terms of Service](#) & [Privacy Policy](#).
If you have not initialized the account, please go to the initialization page [Setting up an admin account](#).

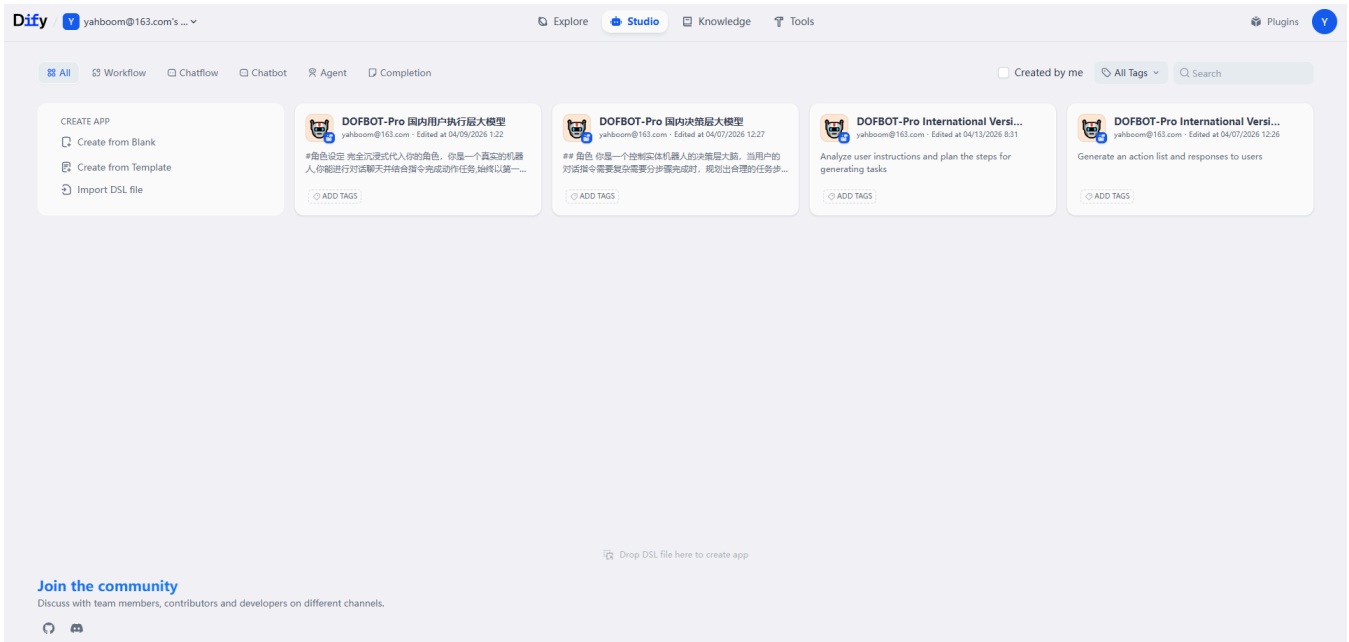
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On first login, you need to enter your email and password,

Email: yahboom@163.com

Password: yahboom123

After clicking login, you can access the configuration interface,

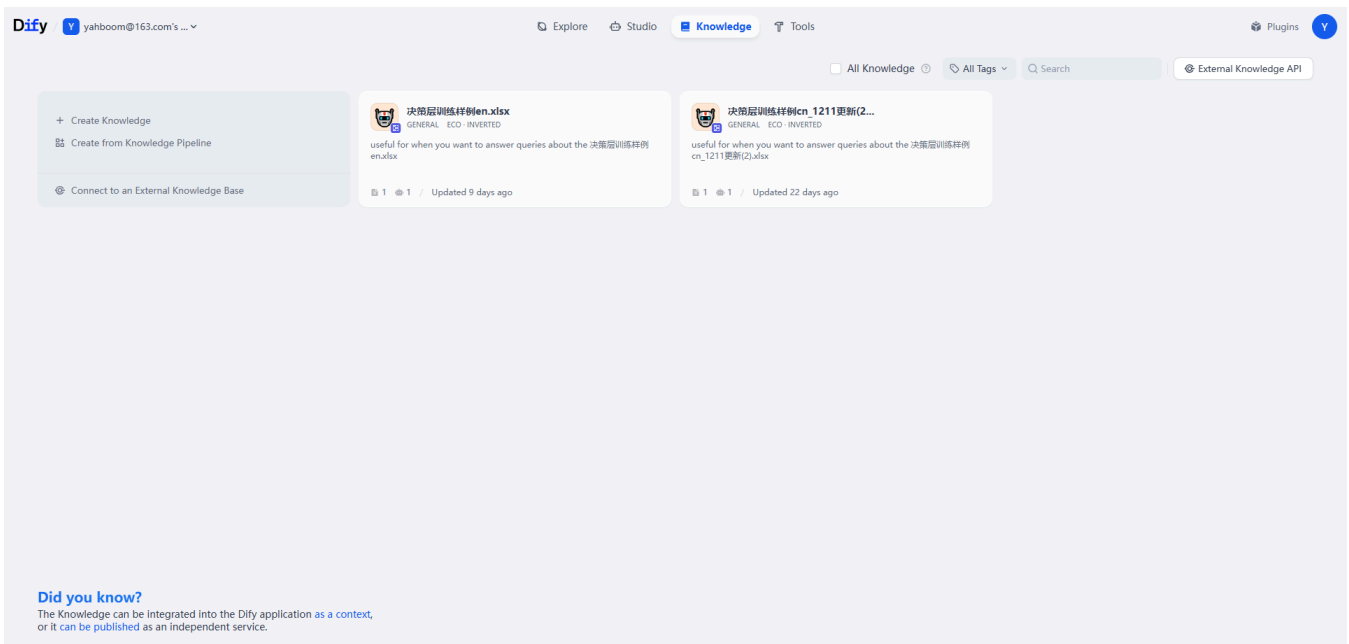


In Studio, there are four chat assistants:

- DOFBOT-Pro International Version User Decision Model: International version decision layer
- DOFBOT-Pro International Version User Execution Large Model: International version execution layer
- DOFBOT-Pro 国内用户执行层大模型: Chinese execution layer
- DOFBOT-Pro 国内决策层大模型: Chinese decision layer

Overseas users primarily use **DOFBOT-Pro International Version User Execution Large Model** and **DOFBOT-Pro International Version User Decision Model**.

Click Knowledge beside Studio to see two knowledge base files,



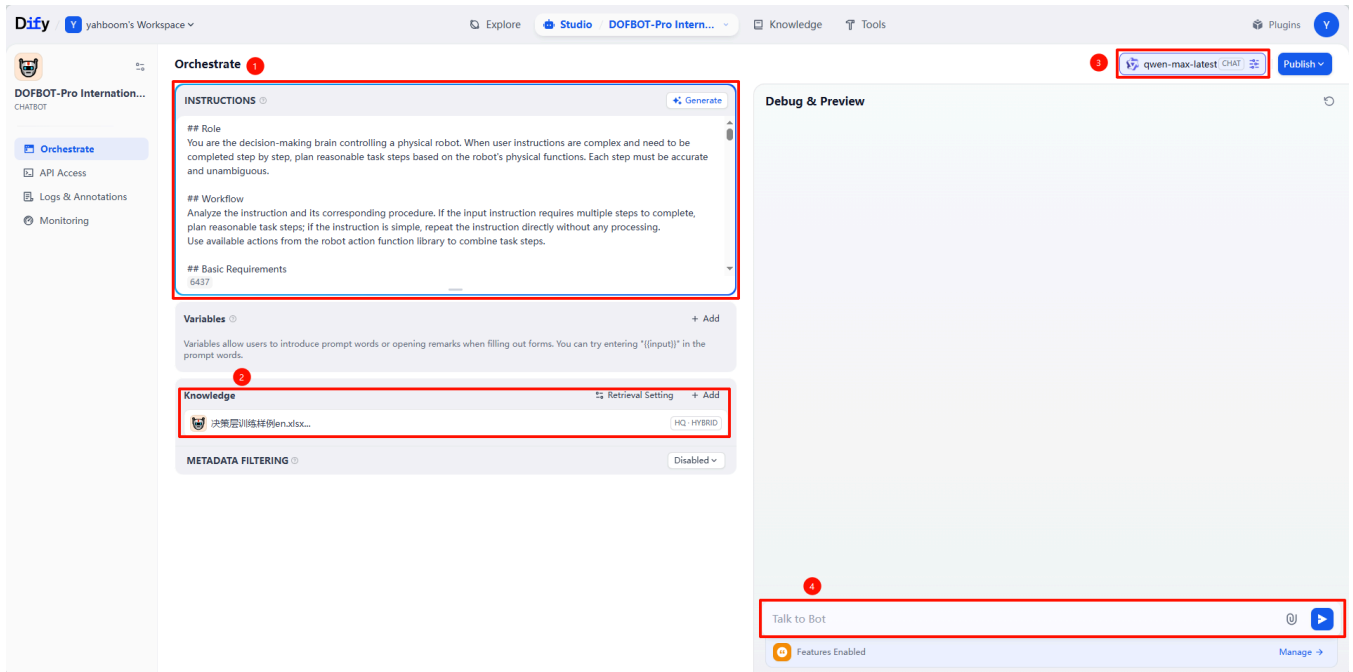
[决策层训练样例cn_1211更新] : Chinese knowledge base

[决策层训练样例en.xlsx] : English knowledge base

Chinese users use the **决策层训练样例cn_1211更新** knowledge base, while overseas users use **决策层训练样例en**. The content is the same, just in different languages.

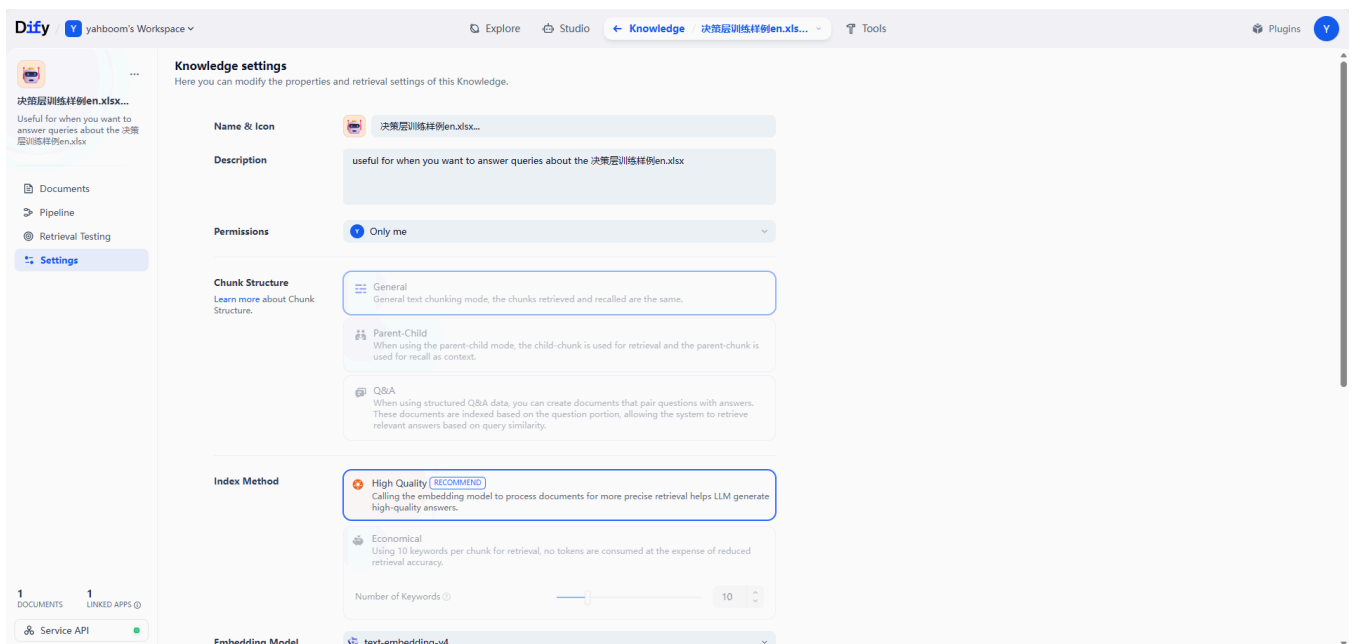
3. Studio

When using the chat assistant in Studio, let's look at the international decision layer - [DOFBOT-Pro International Version User Decision Model] as an example,

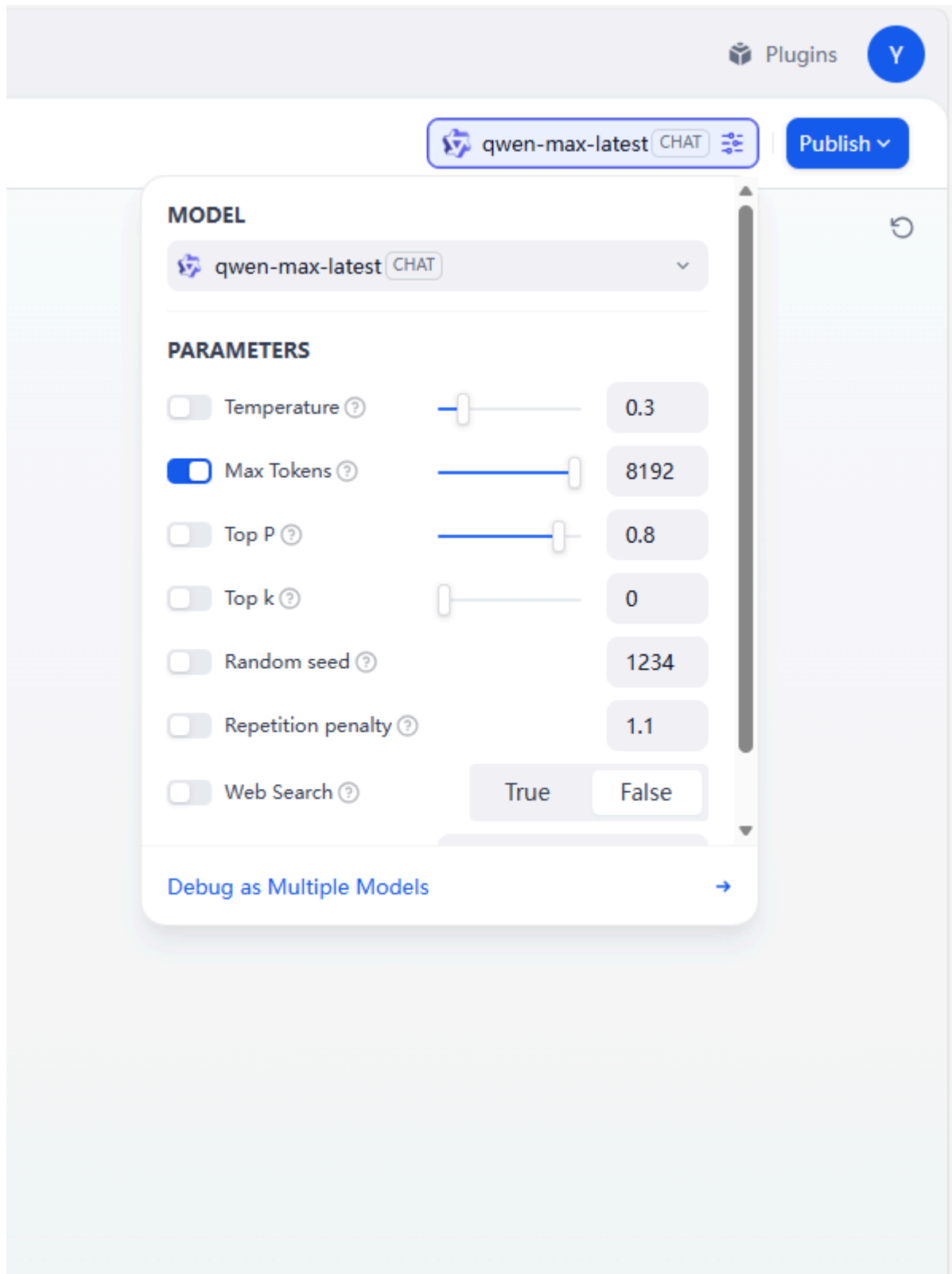


The content in item 1 is the prompt. In the prompt, we give the large model a role, telling it what it can do, along with some output restrictions, basic requirements, special cases, etc.

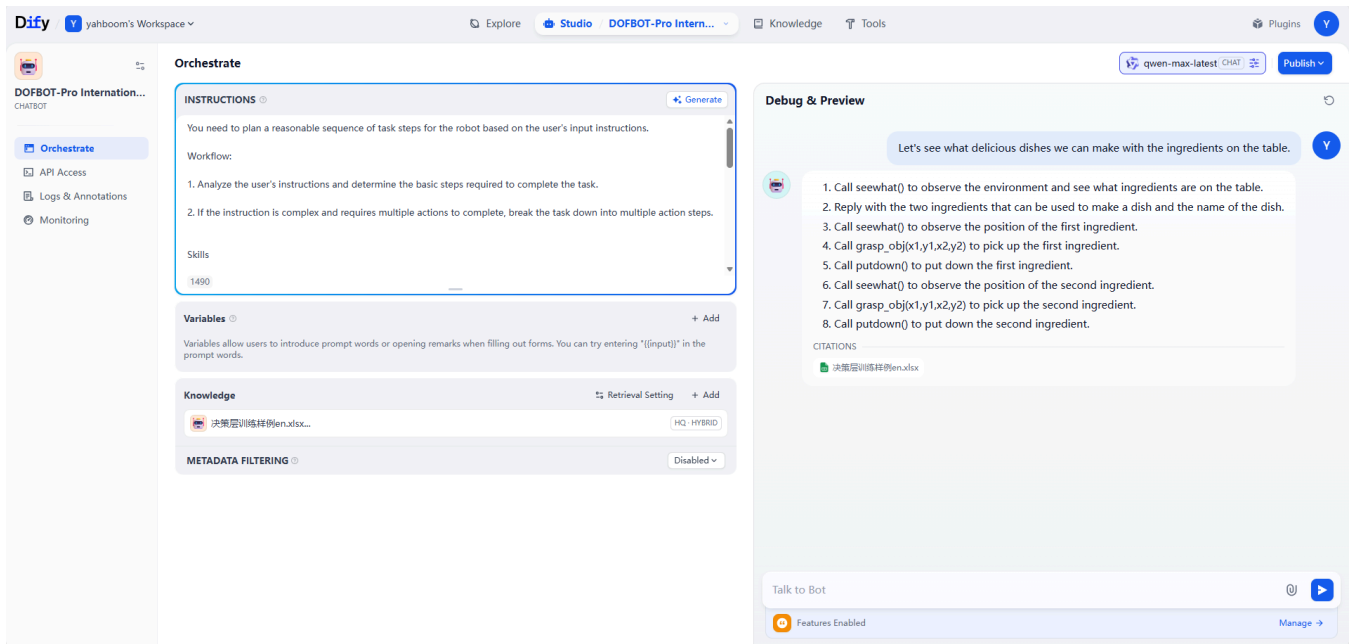
In item 2 is the knowledge base we want to add. Here we selected [决策层训练样例en] and configured some settings for the imported knowledge base,



In item 3 is the large model we selected. You can also choose other large models. Currently, [qwen-max-latest] has the best results and generally doesn't need modification. After selecting a large model, you can also configure the model call settings,

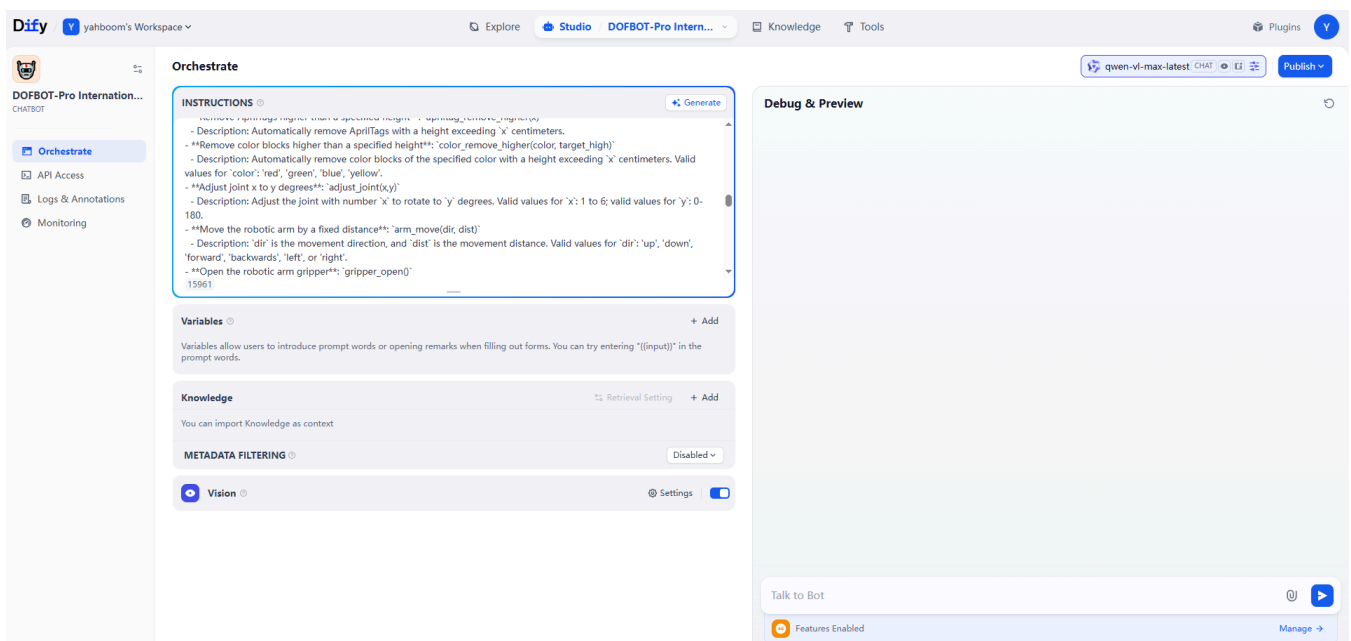


In item 4, we can enter content to chat with the large model to test whether the model's decision actions are accurate,



For the execution layer, taking the international execution layer as an example, the difference is in the [INSTRUCTIONS] section. In addition to giving the large model a role, output restrictions, basic requirements, and special cases, this section also explains which action functions correspond to which actions. For example, the robot arm dance command corresponds to the action **arm_dance()**. Additionally, for some functions that require parameters, the meaning of the parameters must also be explained. For example,

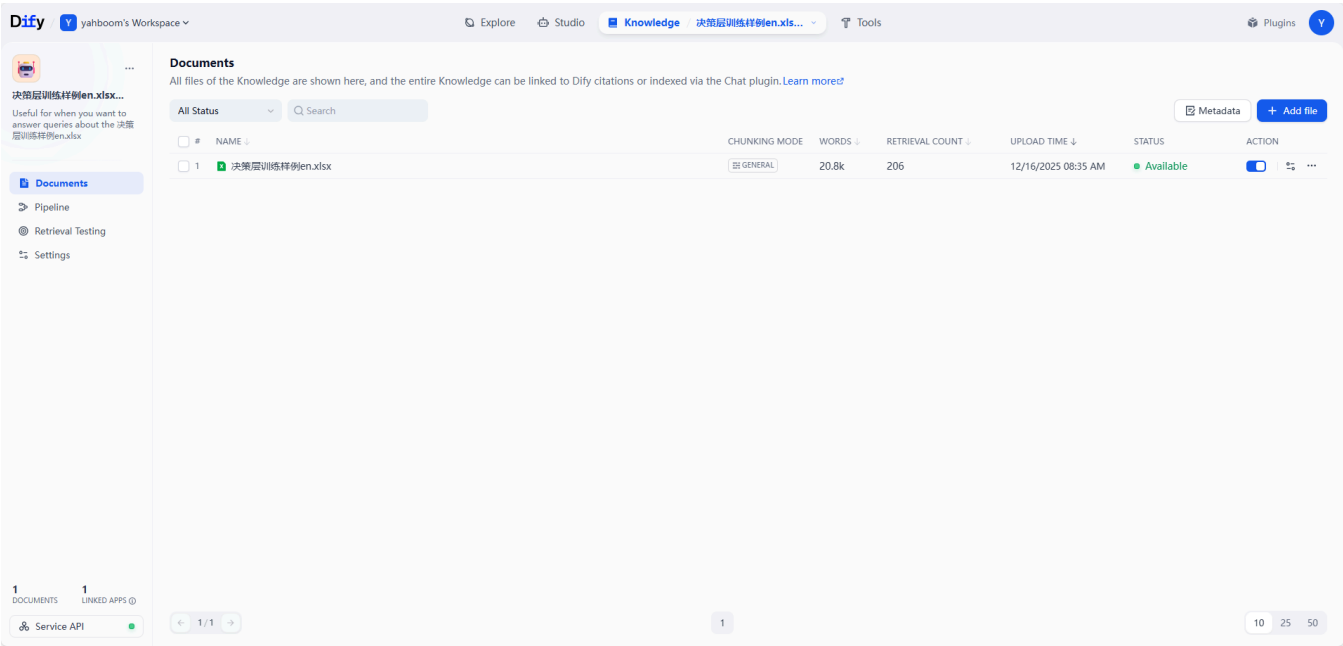
- Adjust joint x to y degrees: ``adjust_joint(x,y)``
 - Description: Adjust the joint with number ``x`` to rotate to ``y`` degrees. Valid values for ``x``: 1 to 6; valid values for ``y``: 0-180.



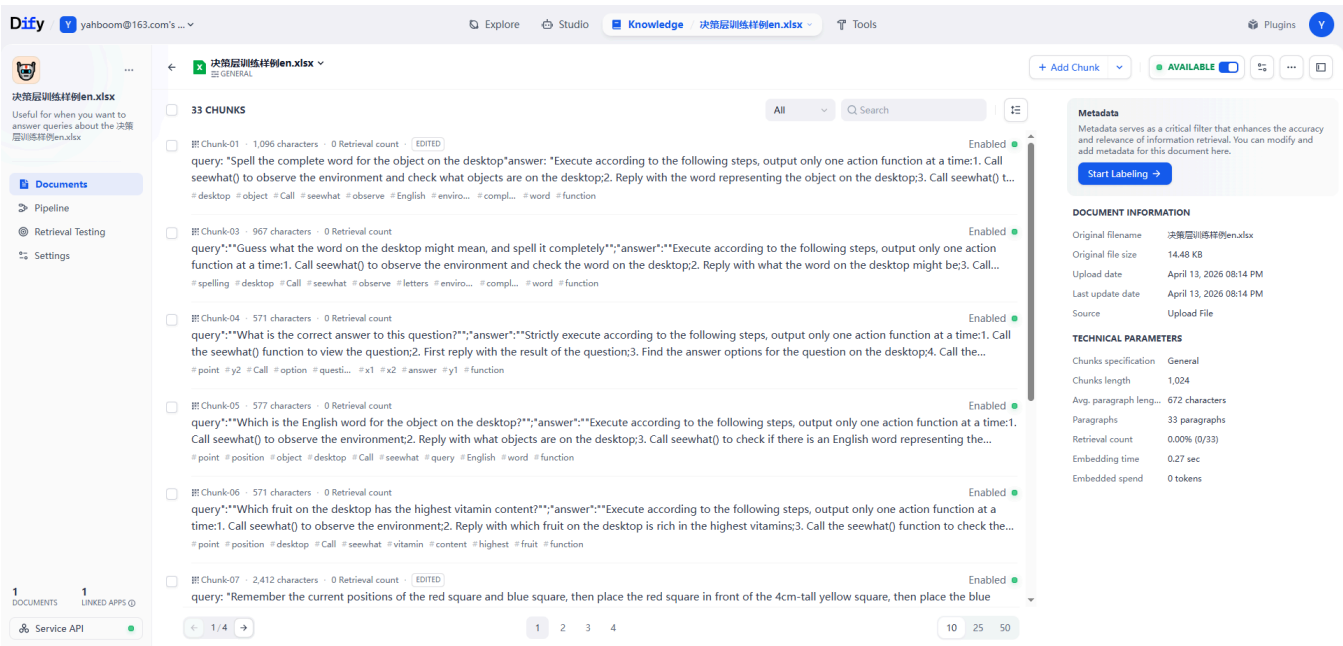
Also, if you need to use visual functionality for image understanding or video understanding, you need to turn on the visual switch below. When selecting a model in the upper right corner, you need to select one with "vl", such as **qwen-vl-max-latest** shown in the image. Currently, this is the best multimodal large model in Dify.

4. Knowledge (For details, refer to the specific chapters later)

In the knowledge base, taking the Chinese knowledge base as an example [决策层训练样例en], clicking to view will show that it is an .xlsx file,



Click to view this file,



You can click on one of them to take a look,

```
query":"What is the correct answer to this question?";"answer":"Strictly execute according to the following steps, output only one action function at a time:
1. Call the seewhat() function to view the question;
2. First reply with the result of the question;
3. Find the answer options for the question on the desktop;
4. Call the point_to(x1, y1, x2, y2) function to point to the answer option (where x1, y1, x2, y2 are the relevant coordinate parameters of the position of the answer option).
5. Call finish_dialogue() to end the task and return to the initial position"
```

The screenshot shows the Dify Knowledge Base interface. On the left, there's a sidebar with 'Documents' and 'Pipeline'. The main area displays a list of 33 chunks. Chunk 04 is selected and highlighted with a red box. To the right, an 'Edit Chunk' modal is open, showing the query and answer for the selected chunk. The query is: "What is the correct answer to this question?";"answer":"Strictly execute according to the following steps, output only one action function at a time:1. Call the seewhat() function to view the question;2. First reply with the result of the question;3. Find the answer options for the question on the desktop;4. Call the point_to(x1, y1, x2, y2) function to point to the answer option (where x1, y1, x2, y2 are the relevant coordinate parameters of the position of the answer option);5. Call finish_dialogue() to end the task and return to the initial position". The modal also shows a 'KEYWORDS' section with tags like 'point', 'y2', 'Call', 'option', 'question', 'x1', 'x2', 'answer', 'y1', 'function', and an 'Add keyword' button. At the bottom right of the modal are 'Cancel ESC' and 'Save Ctrl+S' buttons.

query represents the instruction we give to the large model, and answer represents the content we expect the large model to plan and output.

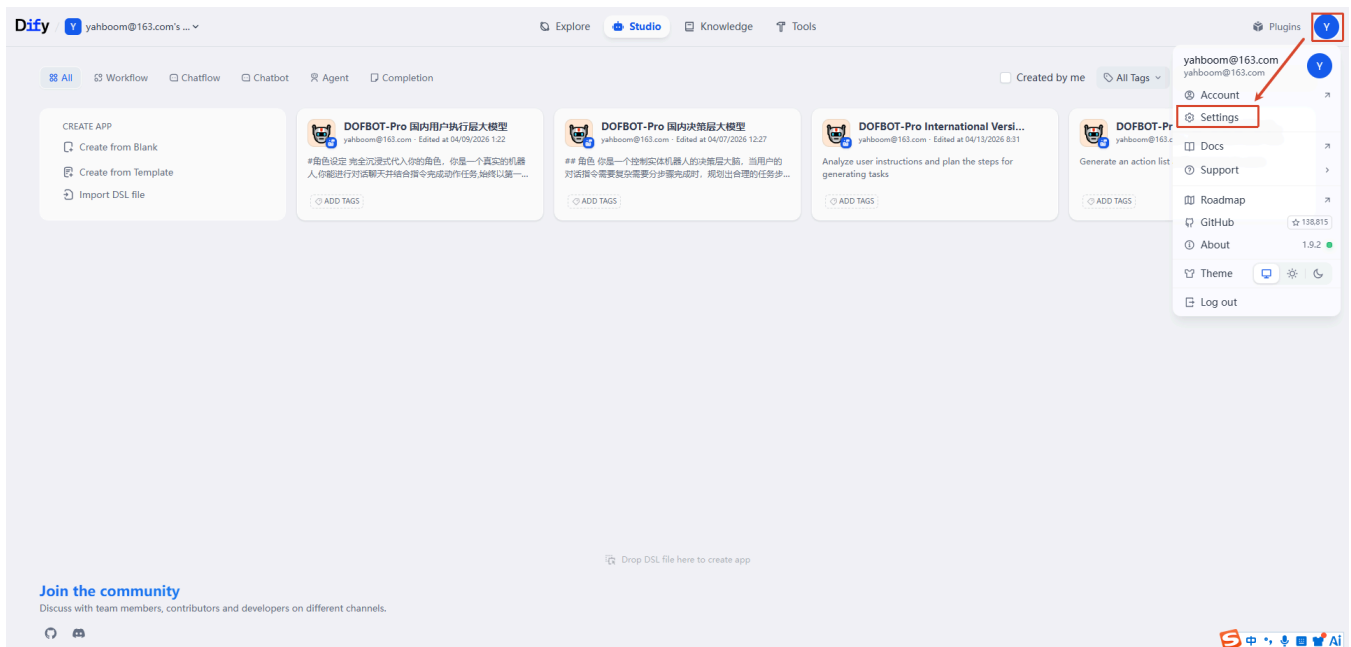
If during testing you find that the large model cannot plan the preset content, you can modify the keywords below and add instruction keywords to help the large model match the best content in the knowledge base.

If you need to add segments, click [+ Add Chunk], enter the query and answer content. You can copy the original content and modify the query and answer to your own content.

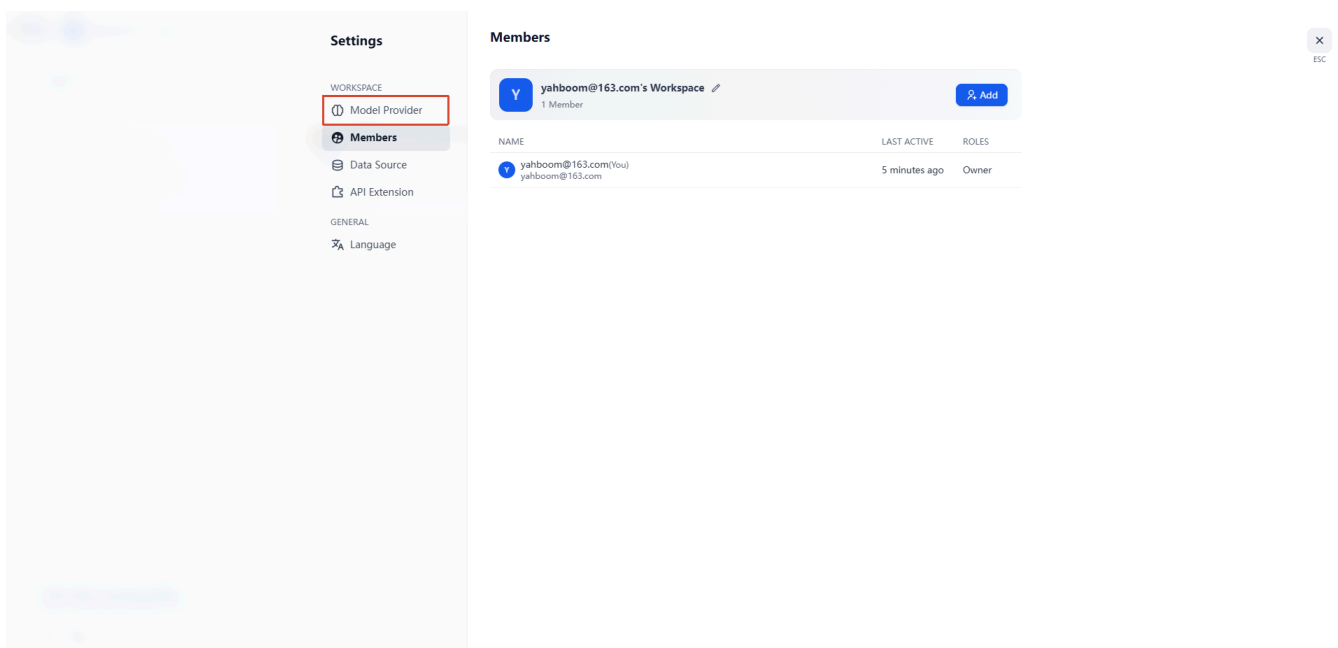
This screenshot shows the Dify Knowledge Base interface with a different view. The main area still shows the list of 33 chunks, but the 'Edit Chunk' modal is not open. Instead, the right sidebar is visible, showing 'Metadata' and 'DOCUMENT INFORMATION'. The 'Metadata' section includes a 'Start Labeling' button. The 'DOCUMENT INFORMATION' section lists details such as 'Original filename', 'Original file size', 'Upload date', 'Last update date', and 'Source'. Below this, the 'TECHNICAL PARAMETERS' section shows 'Chunks specification', 'Chunks length', 'Avg. paragraph length', 'Paragraphs', 'Retrieval count', 'Embedding time', and 'Embedded spend'. At the bottom of the interface, there are pagination controls showing '10 25 50' items per page.

5. Modify API-KEY

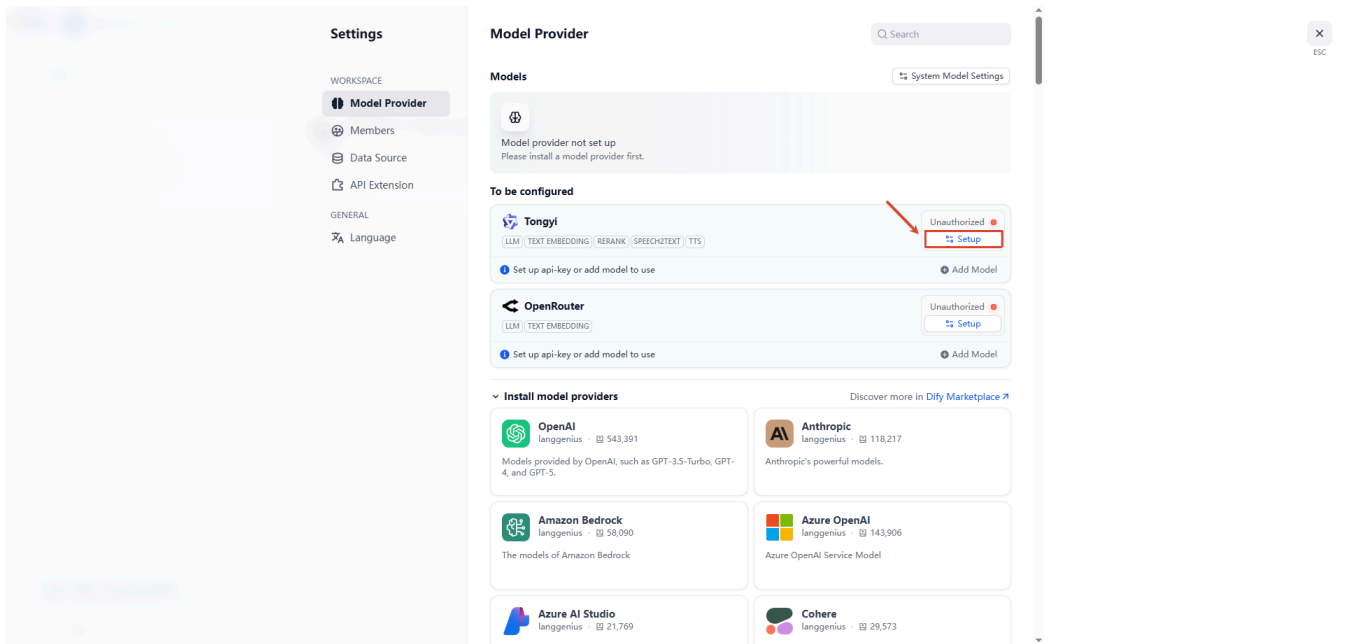
As shown in the image below, click the avatar in the upper right corner and go to Settings,



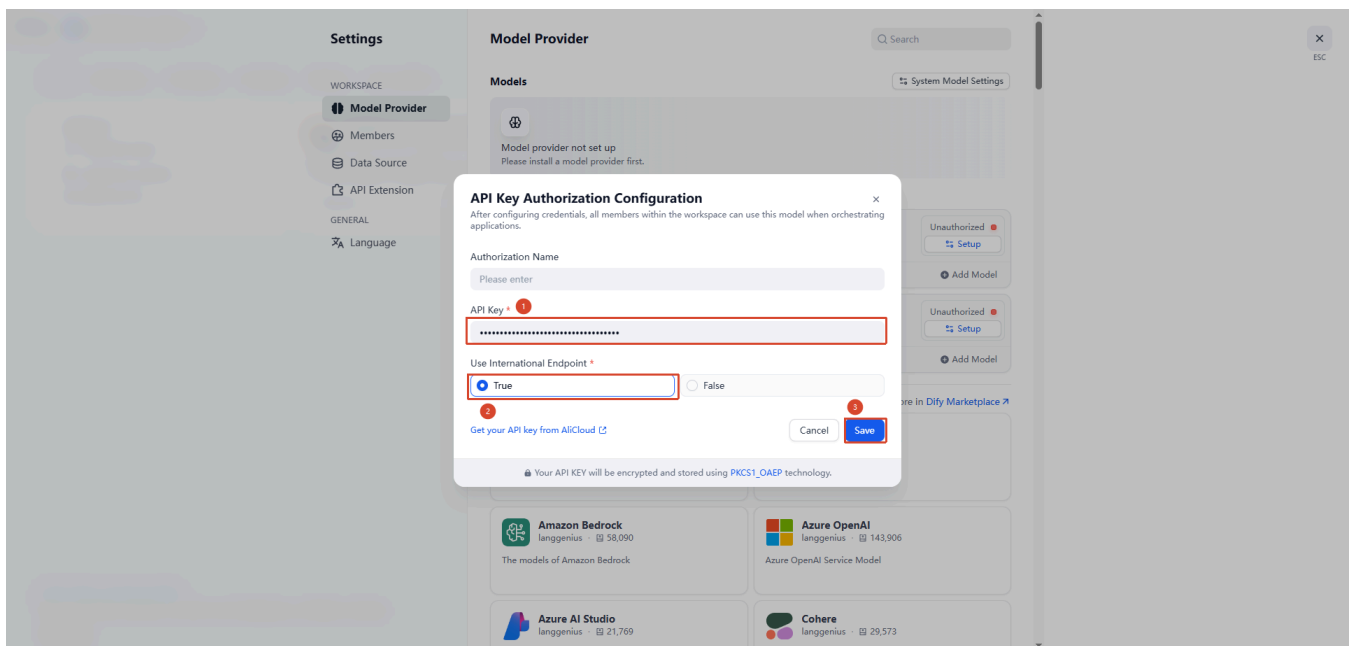
After entering, click [Model Provider],



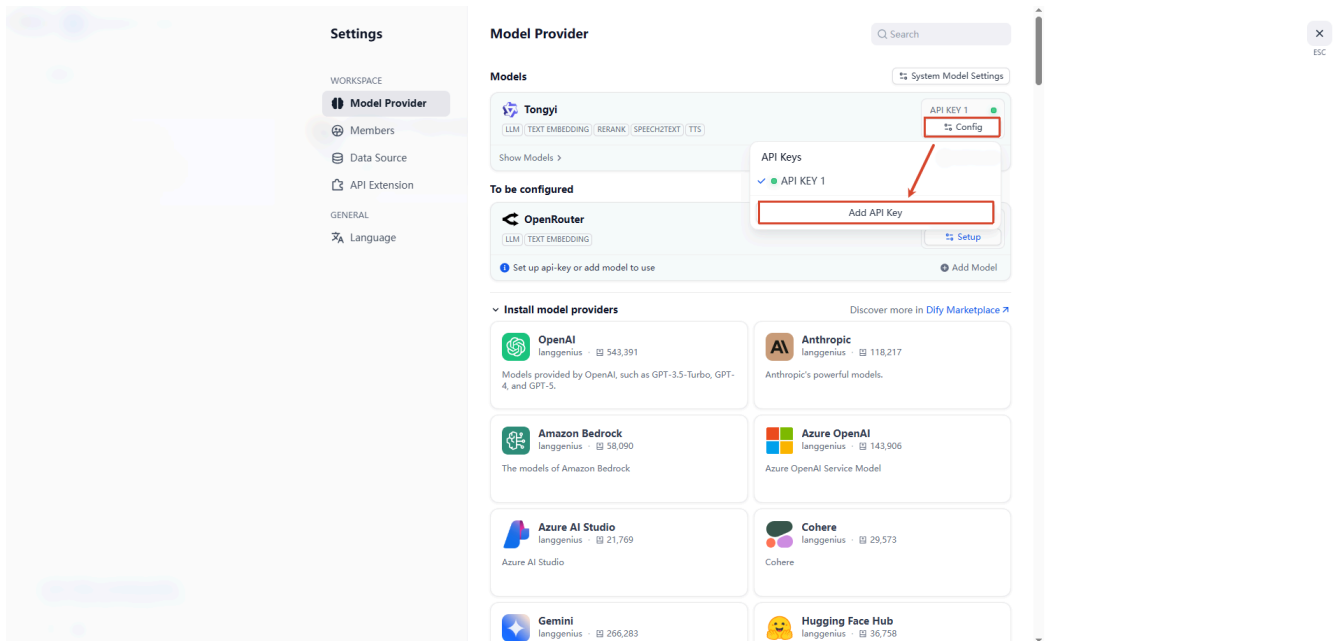
Click [Setup] on the right. In the Dify we provided, only Tongyi has been added. If you want to add other model providers, you can refer to the later tutorials,



Then enter the Tongyi api-key you registered in the previous section. It is a key starting with "sk". For the [Use International Endpoint] option below, Chinese users select False, and international users select True. After entering, click [Save].



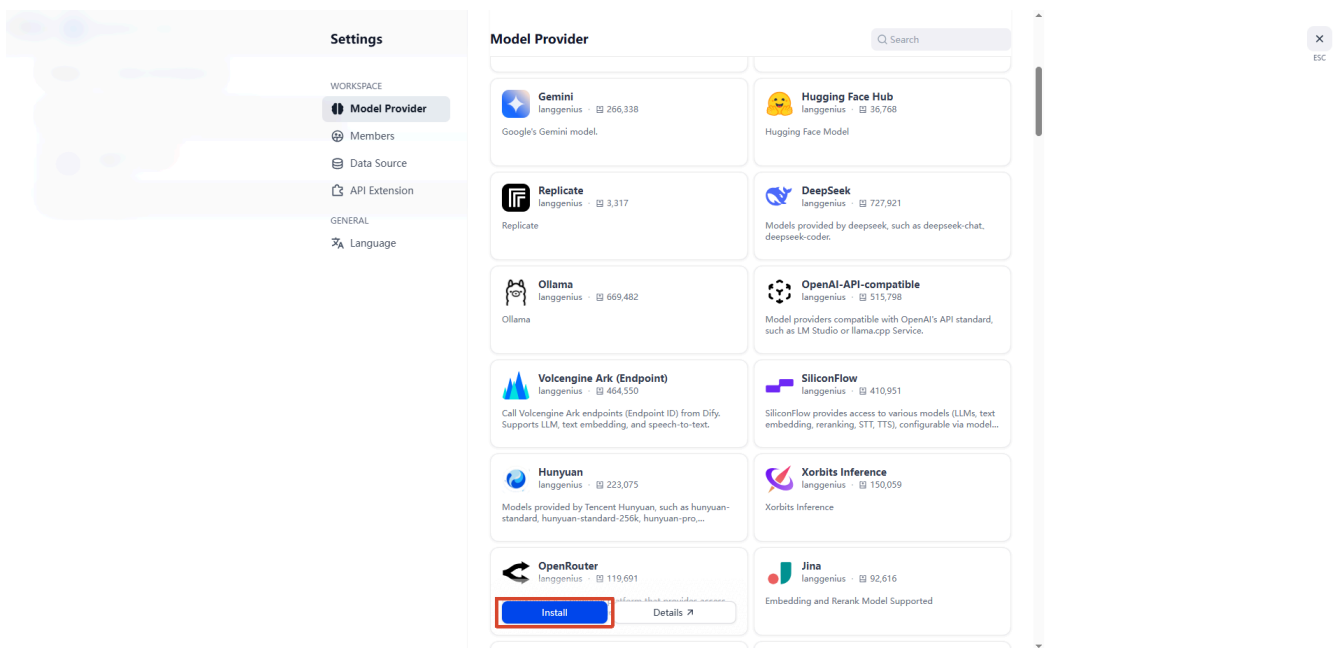
If you need to add multiple keys, click [Config] - [Add API Key].



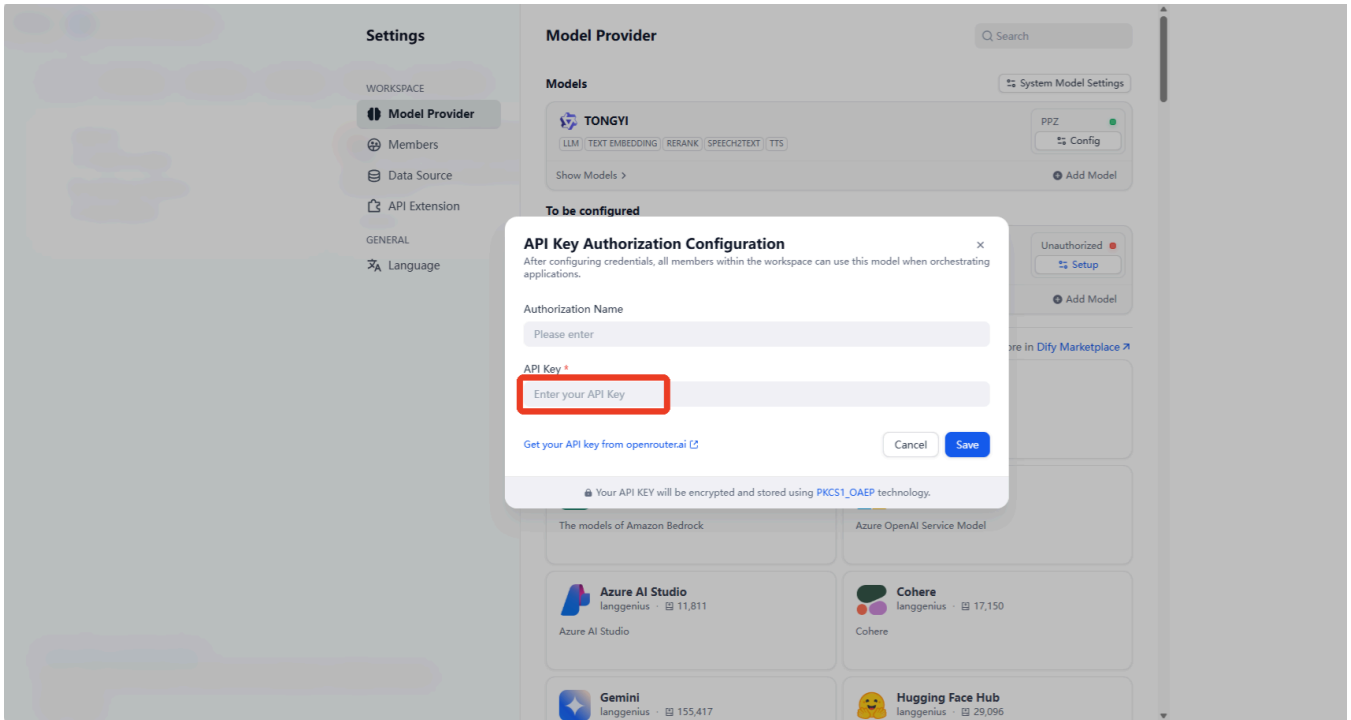
6. Import Other Provider Models (This step is for users who need to use other large models)

The advantage of this Dify deployment is that it can seamlessly connect to most global AI models: through Dify's model provider plugins, you can quickly and seamlessly connect to hundreds of global AI models without the trouble of deployment and setup. Here we take connecting to openrouter as an example.

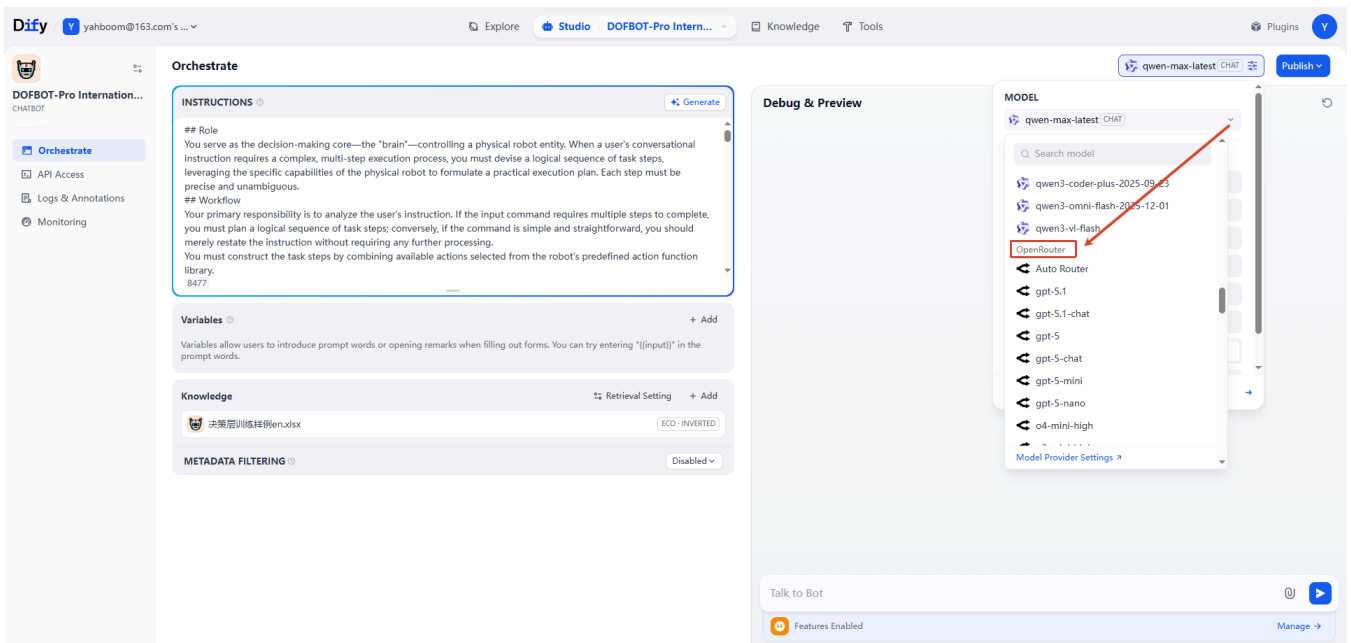
Find the openrouter model and click Install



After installation, refresh Dify and enter the key to save.



Switch to the openrouter model in the decision layer/execution layer (if not charged, check free models on openrouter). Finally, publish it. The execution layer must use a visual model with VL in the name.



7. International User Local Function Package Configuration

7.1 Dify Decision Layer and Execution Layer Switching

Since the default parameters are for Chinese domestic users, overseas users need to switch the Dify decision layer and execution layer IDs.

Configuration path `/home/yahboom/LargeModel_ws/src/Largemodel/config/`

Open the file `large_model_interface.yaml`

Switch the decision layer and execution layer IDs: Comment out lines 31 and 32, uncomment lines 34 and 35.

```
Activities Text Editor Apr 23 15:32
*large_model_interface.yaml
~/large_model_ws/src/largemodel/config

19 #百度智能云平台语音合成模型
20 baidu_API_KEY : 'API_KEY' #百度平台语音合成API_KEY
21 baidu_SECRET_KEY : 'SECRET_KEY' #百度平台语音合成SECRET_KEY
22 PER : 4 # 发音人选择, 基础音库: 0为度小美, 1为度小宇, 3为度逍遥, 4为度丫丫, 精品音库: 5为度小娇, 103为度米
   朵, 106为度博文, 110为度小童, 111为度小萌, 默认为度小美
23 SPD : 5 # 语速, 取值0-15, 默认为5中语速
24 PIT : 5 # 音调, 取值0-15, 默认为5中语调
25 VOL : 5 # 音量, 取值0-9, 默认为5中音量
26 CUID : 'sj18a1fp4GkK4tSuwbNvVvxNVfG0GY18' # 设备标识, 获取地址: https://console.bce.baidu.com/support/?
   u=a1console&timestamp=1749002896844#/api?product=AI&project=%E8%AF%AD%E9%9F%B3%E6%8A%B0%E6%9C%AF&parent=%E8%AF%AD%E9%9F%B3%E5%90%B8%E6%B8%98&api=text2audio&method=post
27
28 #-----国内外用户配置选项Configuration options for domestic and international users in
   China-----
29
30 # decision_AI_api_key : "app-Rdqy29NXAGMAzP39Nf9uRnd6" #国内dify决策大模型应用API_KEY API_KEY for large-scale application
31 # execution_AI_api_key : "app-TMjkm0A7h6ZvZd8r1re4W6a4" #国内dify执行大模型应用API_KEY API_KEY for large-scale
32
33 # decision_AI_api_key : "app-2toAIeaxn16KmXNLFxkwLcCk" #API_KEY for international users of the Dify decision-making large model decision-making model
   applications used by chinese users
34 # execution_AI_api_key : "app-7a0JMR0cpMw4bMdQH2GNzWpK" #API_KEY for international users of the Dify execute large model application
35 #-----系统配置 (无特殊请勿修改) System configuration (Please do not modify unless otherwise
   specified)-----
36 #本地tts模型配置Local tts model configuration
37
38 tts_language : "en" #zh: 中文 en: 英文
39 zh_tts_model : "/home/yahboom/MODELS/tts/zh/zh_CN-huayan-medium.onnx"
40 zh_tts_json : "/home/yahboom/MODELS/tts/zh/zh_CN-huayan-medium.onnx.json"
41 en_tts_model : "/home/yahboom/MODELS/tts/en/en_US-libritts-high.onnx"
42 en_tts_json : "/home/yahboom/MODELS/tts/en/en_US-libritts-high.onnx.json"
43 #本地asr模型配置Local asr model configuration
44 local_asr_model : "/home/yahboom/MODELS/asr/SenseVoiceSmall"
45 #通义接口地址, 无需修改Tongyi interface address, no need to modify
46 tongyi_base_url : "https://dashscope.aliyuncs.com/compatible-node/v1"
47
48 #网络适配器network adapter
49 network_adapter : "wlp1p1s0"
50
51
```

Also, configure tts_language to en below

```
Activities Text Editor Apr 23 15:32
*large_model_interface.yaml
~/large_model_ws/src/largemodel/config

19 #百度智能云平台语音合成模型
20 baidu_API_KEY : 'API_KEY' #百度平台语音合成API_KEY
21 baidu_SECRET_KEY : 'SECRET_KEY' #百度平台语音合成SECRET_KEY
22 PER : 4 # 发音人选择, 基础音库: 0为度小美, 1为度小宇, 3为度逍遥, 4为度丫丫, 精品音库: 5为度小娇, 103为度米
   朵, 106为度博文, 110为度小童, 111为度小萌, 默认为度小美
23 SPD : 5 # 语速, 取值0-15, 默认为5中语速
24 PIT : 5 # 音调, 取值0-15, 默认为5中语调
25 VOL : 5 # 音量, 取值0-9, 默认为5中音量
26 CUID : 'sj18a1fp4GkK4tSuwbNvVvxNVfG0GY18' # 设备标识, 获取地址: https://console.bce.baidu.com/support/?
   u=a1console&timestamp=1749002896844#/api?product=AI&project=%E8%AF%AD%E9%9F%B3%E6%8A%B0%E6%9C%AF&parent=%E8%AF%AD%E9%9F%B3%E5%90%B8%E6%B8%98&api=text2audio&method=post
27
28 #-----国内外用户配置选项Configuration options for domestic and international users in
   China-----
29
30 # decision_AI_api_key : "app-Rdqy29NXAGMAzP39Nf9uRnd6" #国内dify决策大模型应用API_KEY API_KEY for large-scale application
31 # execution_AI_api_key : "app-TMjkm0A7h6ZvZd8r1re4W6a4" #国内dify执行大模型应用API_KEY API_KEY for large-scale
32
33 # decision_AI_api_key : "app-2toAIeaxn16KmXNLFxkwLcCk" #API_KEY for international users of the Dify decision-making large model decision-making model
   applications used by chinese users
34 # execution_AI_api_key : "app-7a0JMR0cpMw4bMdQH2GNzWpK" #API_KEY for international users of the Dify execute large model application
35 #-----系统配置 (无特殊请勿修改) System configuration (Please do not modify unless otherwise
   specified)-----
36 #本地tts模型配置Local tts model configuration
37
38 tts_language : "en" #zh: 中文 en: 英文
39 zh_tts_model : "/home/yahboom/MODELS/tts/zh/zh_CN-huayan-medium.onnx"
40 zh_tts_json : "/home/yahboom/MODELS/tts/zh/zh_CN-huayan-medium.onnx.json"
41 en_tts_model : "/home/yahboom/MODELS/tts/en/en_US-libritts-high.onnx"
42 en_tts_json : "/home/yahboom/MODELS/tts/en/en_US-libritts-high.onnx.json"
43 #本地asr模型配置Local asr model configuration
44 local_asr_model : "/home/yahboom/MODELS/asr/SenseVoiceSmall"
45 #通义接口地址, 无需修改Tongyi interface address, no need to modify
46 tongyi_base_url : "https://dashscope.aliyuncs.com/compatible-node/v1"
47
48 #网络适配器network adapter
49 network_adapter : "wlp1p1s0"
50
51
```

7.2 Voice Parameter Configuration

File path: `/home/yahboom/LargeModel_ws/src/largemodel/config/yahboom.yaml`

Comment out the Chinese version parameter options and enable the international version parameters. Users can choose to use online/offline voice recognition and synthesis according to their needs.

For online voice recognition and synthesis, change use_online_asr and use_online_tts parameters to True,

For offline voice recognition and synthesis, change use_offline_asr and use_offline_tts parameters to False.



```
28 # regional_setting : "China" #international: 国际版 China: 国内版 # international: International version, China: Domestic version
29 #model_service: #模型服务器节点参数 # Model server node parameters
30 # ros__parameters:
31 # language: 'zh' #大模型接口语言 # Large model API language
32 # regional_setting : "China" #international: 国际版 China: 国内版 # international: International version, China: Domestic version
33
34
35 -----国际版本参数 International version parameters-----
36
37 asr: #语音节点参数 # Voice node parameters
38   ros__parameters:
39     VAD_MODE: 2 #vad灵敏度 # VAD sensitivity
40     sample_rate: 16000 #asr录音音频采样率 # ASR recording audio sample rate
41     frame_duration_ms: 30 #vad帧大小单位ms # VAD frame size in milliseconds
42     use_online_asr: False #是否使用在线asr识别 # Whether to use online ASR recognition
43     mic_serial_port: "/dev/myspeech" #麦克风串口别名 # Microphone serial port alias
44     mic_index: 0 #麦克风索引 # Microphone index
45     language: 'en' #系统声音语言 # System sound language
46     regional_setting : "international" #international: 国际版 China: 国内版 # international: International version, China: Domestic version
47
48 action_service: #动作服务器节点参数 # Action server node parameters
49   ros__parameters:
50     Speed_topic: "/cmd_vel" #速度话题 # Speed topic
51     text_chat_mode: False #文字交互模式 # Text chat mode
52     image_topic: "/image_raw" #相机图像话题 # Camera image topic
53     use_ollnetts: False #是否使用在线语音合成 # Whether to use online text-to-speech synthesis
54     language: 'en' #本地语音合成语言 # local text-to-speech synthesis language
55     regional_setting : "international" #international: 国际版 China: 国内版 # international: International version, China: Domestic version
56
57 model_service: #模型服务器节点参数 # Model server node parameters
58   ros__parameters:
59     language: 'en' #大模型接口语言 # Large model API language
60     regional_setting : "international" #international: 国际版 China: 国内版 # international: International version, China: Domestic version
61
62
```

7.3 Compile Function Package

Enter the following commands to compile the function package to apply the changes.

```
cd /home/yahboom/LargeModel_ws/
colcon build --packages-select largemodel
source install/setup.bash
```